MIL-C-36229(DM)
1 July 1964

MILITARY SPECIFICATION

CYLINDERS, GRADUATED, LABORATORY, POLYPROPYLENE

1. SCOPE

1.1 Scope. This specification covers polypropylene, graduated cylinders, suitable for laboratory use.

1.2 Classification. Polypropylene graduated cylinders shall be of the following sizes, as specified (see 6.1):

Size 1. - 10 milliliters (ml.) nominal capacity.
Size 2. - 25 ml. nominal capacity.
Size 3. - 50 ml. nominal capacity.
Size 4. - 100 ml. nominal capacity.
Size 5. - 250 ml. nominal capacity.
Size 6. - 500 ml. nominal capacity.
Size 7. - 1000 ml. nominal capacity.
Size 8. - 2000 ml. nominal capacity.

FSC 6640

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T08:15Z
Check the source to verify that this is the current version before use.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-B-566	Boxes, Folding, Paperboard.
PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Box, Fiberboard.
PPP-B-676	Boxes, Set-Up, Paperboard.
PPP-T-76	Tape, Pressure-Sensitive Adhesive Paper, Water Resistant (for Carton Sealing).

Military

MIL-L-10547	Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Source: https://assist.dla.mil -- Downloaded: 202640-21T08:15Z
Check the source to verify that this is the current version before use.

2.2 Other publication. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

U. S. DEPARTMENT OF COMMERCE - NATIONAL BUREAU OF STANDARDS

NBS Circular 602 Testing of Glass Volumetric Apparatus.

(Application for copies, accompanied by money order, coupon, or check, should be addressed to the Superintendent of Documents, Government Printing Office, Washington 25, D. C. Price may be obtained from the Superintendent of Documents.)

3. REQUIREMENTS

3.1 Material. Material shall be good-quality translucent polypropylene, suitable for the purpose intended.

3.2 Construction. Cylinders shall be rigid, and shall be provided with a barrel and base.

3.2.1 Barrel. The barrel shall be cylindrical, and shall be provided with a suitable pouring lip. Details of design shall be as shown in table I.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T08:15Z
Check the source to verify that this is the current version before use.

Table I. Design Details for Barrel

Size	Distance from top of scale to top of barrel (centimeters)		Minimum wall thickness (millimeters)	Maximum draft (taper)
	Maximum	Minimum		
1	5.0	2.0	0.80	0° 15'
2	8.0	2.5	1.00	1° 00'
3	8.0	2.5	1.60	0° 30'
4	8.0	3.0	1.60	0° 30'
5	10.0	4.0	1.75	0° 15'
6	10.0	4.5	1.75	0° 20'
7	10.0	5.0	1.75	1° 00'
8	10.0	5.0	1.75	1° 00'

3.2.2 Base. The base shall be octagonal, perpendicular to the axis of the cylinder, and shall be of such dimension that the empty cylinder shall not fall when tested as specified in 4.4.1.

3.2.3 Calibration. Cylinders shall be calibrated to deliver the indicated volume at 25° centigrade (C.) (77°) Fahrenheit (F.).

3.2.4 Scale. Cylinders shall have a single scale indicated in ml.

3.2.5 Graduated intervals. The graduation ranges and intervals shall be as shown in table II. The values of all intervals shall be the same and shall be as indicated (± percent), when tested as specified in 4.4.2.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T08:15Z
Check the source to verify that this is the current version before use.

Table II. Graduation Intervals

Size	To be graduated between (inclusive)	Graduations			Minimum width of intervals (millimeters)
		Least	Intermediate	Main	
1	1 to 10	-	0.5	1	1.0
2	5 to 25	0.5	1	5	0.9
3	10 to 50	-	1	10	1.8
4	10 to 100	1	5	10	1.2
5	50 to 250	-	10	50	1.5
6	100 to 500	5	25	100	2.4
7	100 to 1000	-	10	100	2.7
8	200 to 2000	20	100	200	3.5

Note: Additional graduations may be added provided they meet the requirements of this specification.

3.2.6 Graduation markings. Graduation markings shall be raised, permanent, legible, and a maximum of 0.6 millimeter wide. Graduation lines shall be perpendicular to the vertical axis of the barrel. Main graduations shall be marked, and intermediate and least graduations shall not be marked. When additional graduations are placed beneath the lowest main graduation, initial graduation line may be marked. Main graduations shall extend a minimum of one-third of the way around the cylinders, intermediate graduations a minimum of one-fourth, and the least graduations a minimum of one-sixth.

3.3 Heat resistance. The cylinder shall not soften, deform, crack, or become opaque, when tested as specified in 4.4.3.

3.4 Resistance to dripping. The exterior of the cylinder shall show no evidence of water, when the cylinder is tested as specified in 4.4.4.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T08:15Z
Check the source to verify that this is the current version before use.

3.5 Identification marking. Cylinders shall be permanently and legibly marked in raised lettering with the name or registered trademark of the manufacturer, calibration means (to deliver), and temperature (25 degrees centigrade). Calibration means and temperature may be abbreviated.

3.6 Workmanship. Workmanship shall be first class throughout. Cylinders shall be free from defects which detract from their appearance or may impair serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual, auditory, etc., investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.2 Certificates of quality. When available, certificates of quality, supplied by the manufacturer of the component or material, listing the specified test method and test results obtained, may be furnished in lieu of actual performance of such testing by the contractor.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with Standard MIL-STD-105.

	Inspection level	AQL (percent defective)
For visual examination		
Major defects	II	2.5
Minor defects	II	4.0
For dimensional examination	S-1	2.5

Source: https://assist.dla.mil -- Downloaded: 202610-21T08:15Z
Check the source to verify that this is the current version before use.

4.2.2 For tests. Sampling for tests shall be conducted in accordance with Standard MIL-STD-105. Unit of product shall be the cylinder.

Characteristic	Requirement	Test procedure	Inspection level	AQL
Stability	3.2.2	4.4.1	S-1	*
Capacity	3.2.5	4.4.2	S-1	*
Heat resistance	3.3	4.4.3	S-1	*
Resistance to dripping	3.4	4.4.4	S-1	*

* Acceptance number shall be zero.

4.2.3 Where possible, the same sample unit shall be used for the determination of two or more test characteristics.

4.3 Examination. The cylinders shall be examined to determine compliance with all requirements contained in this specification.

4.3.1 Classification of defects. Examination shall be conducted in accordance with the following classification of defects:

Categories and defects*

Major

101 Graduation and calibration markings not correct, legible, or permanent.
102 Cylinder not free of deformed areas.
103 Cylinder not free of opaque areas.
104 Cylinder not free of tears.
105 Cylinder not free of cracks.
106 Cylinder not free of sharp edges.
107 Interior of cylinder not smooth.

Minor

201 Cylinder not free of crazing.
202 Edges of cylinder not free of feathering.
203 Exterior of cylinder not smooth (except for markings and mold lines).
204 Manufacturer's identification not complete, correct, legible, or permanent.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T08:15Z
Check the source to verify that this is the current version before use.

Dimensional

301 Dimension not within tolerance.

* Examination shall not be restricted to the classified possible defects listed above.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Stability. The cylinder shall be placed on a smooth surface which makes a 15° angle with the horizontal.

4.4.2 Capacity. A minimum of three graduations, proportionally spaced, shall be tested in accordance with the method outlined in NBS Circular 602.

4.4.3 Heat resistance. The cylinder shall be placed in an autoclave for a minimum of 20 minutes at 15 pounds per square inch and 121° C. (250° F.).

4.4.4 Resistance to dripping. Cylinder shall be filled with water to the nominal capacity and the water poured out through the pouring lip. (Care should be taken to prevent water from spilling on the exterior of the cylinder during the filling process.)

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Unit of issue. One (1) cylinder, of the size specified, constitutes one unit of issue.

5.1.2 Packaging quantities. The number of units of issue indicated in table III shall be packaged in each unit, intermediate, and exterior container, as applicable, for the required level of protection specified in the procurement document.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T08:15Z
Check the source to verify that this is the current version before use.

Table III. Packaging Quantities

t package	Intermediate package	Exterior container
unit	18 units	216 units
unit	18 units	216 units
unit	18 units	216 units
unit	12 units	144 units
unit	12 units	72 units
unit	8 units	48 units
unit	6 units	36 units
unit	6 units	24 units

1 Packing variation permitted. If the required number of units to be ess than the number of units specified to be overpacked in an exterior such units shall be packed in an exterior container of suitable size and ole to a common carrier, which shall insure safe delivery to

Level A.

1 Unit package. Each unit shall be packaged in a carton of appropriate ucted in accordance with Specification PPP-B-566, PPP-B-636, or 5. As an alternate, for sizes 7 and 8, carton fabricated from 0.030-inch-kraft paperboard may be used. Closure shall be adequate to prevent contents under normal handling. Commercial colors shall be acceptable.

2 Intermediate package. Intermediate package shall be a carton of e size constructed in accordance with specification PPP-B-566, 6, or PPP-B-676. Closure shall be adequate to prevent spillage of ıder normal handling. Commercial colors shall be acceptable.

'acking.

Level A.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T08:15Z
Check the source to verify that this is the current version before use.

5.2.1.1 Exterior container. Exterior container shall be designed for type 2 load, and constructed in accordance with Specification PPP-B-585, style 3, for class 3 use; PPP-B-601, table II, using type III, class 1 plywood; PPP-B-621, table III; or PPP-B-636, type I, class 2.

5.2.1.2 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to Specification MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes.

5.2.1.3 Closure. Closure of wood boxes shall be in accordance with appendix of applicable box specification. Closure of each fiberboard box shall be as specified in the appendix of Specification PPP-B-636. In addition, all seams, including manufacturer's joint, shall be closed by tape conforming to the requirements of Specification PPP-T-76. The tape shall be not less than 2 inches wide. Three strips shall be applied to both top and bottom so that all seams and joints shall be covered their full length. Tape applied over the lengthwise seams shall extend a minimum of 2 inches onto each of the end panels. Tape applied over the end edge seams shall extend a minimum of 2 inches along the side and top, and side and bottom panels.

5.2.1.4 Strapping. Strapping, when required, shall be in accordance with appendix of applicable box specification.

NOTE: Strapping shall not be required for shipments forwarded to a receiving activity within the continental limits of the United States for storage and redistribution.

5.2.2 Level B.

5.2.2.1 Exterior container. Exterior container shall conform to Specification PPP-B-601, table I, using type III, class 1 plywood; or shall be designed for type 2 load, and constructed in accordance with Specification PPP-B-585, style 3, for class 2 use; PPP-B-621, table I; or PPP-B-636, type I, class 1.

5.3 Marking.

5.3.1 Interior packages. Each unit package and each intermediate package shall be marked in accordance with Standard MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable carton is fabricated of water-resistant material. Contract or purchase order number and name of contractor shall be shown.

5.3.2 Exterior container. Exterior container shall be marked in accordance with Standard MIL-STD-129. In addition, the letter "L" shall precede the Federal stock number. One space shall be allowed between the letter "L" and the first digit of the stock number. The terms "Stock number" or "FSN" shall not be indicated.

Source: https://assist.dla.mil -- Downloaded: 202610-21T08:15Z
Check the source to verify that this is the currentversion before use.

ES

ering data. Procurement documents should specify the following:

Number and date of this specification.

Size required (see 1.2).

Selection of applicable level of packing (see 5.2).

s specification does not include all types, classes, grades, sizes, etc.,
odity indicated by the title of this specification, or which are
y available, but is intended to cover the types, etc., which are
ocured to meet Military requirements.

- MD
BuMed
ce. - AFMSG

Preparing activity:
Defense Supply Agency. - DM

- MD
BuMed
ce. - AFMSG

Review information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T08:15Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION
MIL-C-36229(DM) CYLINDERS, GRADUATED, LABORATORY, POLYPROPYLENE

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE PLATE NO. 19419

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T08:15Z
Check the source to verify that this is the current version before use.

FOLD

DEFENSE SUPPLY AGENCY
DEFENSE MEDICAL SUPPLY CENTER
3RD AVENUE AND 29TH STREET
BROOKLYN, N.Y., 11232

OFFICIAL BUSINESS

POSTAGE FEES PAID
DEFENSE SUPPLY AGENCY

Defense Supply Agency
Defense Medical Supply Center
ATTN: Directorate of Technical Operations (Code T-M5)
3rd Avenue and 29th Street
Brooklyn, N.Y., 11232

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T08:15Z
Check the source to verify that this is the current version before use.